

AYUSH VELHAL

ayush.velhal@gmail.com • (469) 763-5767 • [Portfolio](#) • [Github](#) • Dallas/Fort Worth, TX

EDUCATION

University of Texas at Dallas | Bachelor of Science in Computer Science, Minor in Business Administration

Expected May 2027

CERTIFICATIONS

CompTIA Security+ (SY0-701) - Certified
AWS Certified Cloud Practitioner - Certified
Google Cybersecurity Professional Certificate
Microsoft Cybersecurity Analyst Professional Certificate

SKILLS

Programming: Python, Java, SQL
Cybersecurity: Threat Detection, Security Monitoring, Incident Response, SIEM, SOAR, Threat Intelligence, MITRE ATT&CK, OWASP Top 10, IAM, Vulnerability Management
Network Security: TCP/IP, DNS, HTTP/HTTPS, Firewalls, IDS/IPS, Packet Analysis, Network Traffic Analysis
Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda, IAM), Microsoft Azure, Cloud Security, Linux/Unix, Docker, Nginx, Cloudflare
Tools: ELK Stack, Splunk, Wazuh, Wireshark, PostgreSQL, Git, GitHub, Postman
Security Operations: Log Analysis, Alert Triage, Incident Investigation, Security Event Correlation, Malware Analysis, Phishing Analysis

PROJECTS

NovaSOC | Enterprise SOC, SIEM & SOAR Platform | Python, PostgreSQL, AWS EC2, ELK Stack, Docker, SQLAlchemy, Plotly

05/2026

GitHub • [Live Site](#) • [Demo Video](#)

- Architected and deployed an **Enterprise SOC, SIEM & SOAR platform** on AWS EC2 capable of detecting and monitoring hundreds of simulated cyberattacks, including **SSH Brute Force, SQL Injection, DDoS, Malware Activity, Privilege Escalation, and Port Scanning**.
- Built a **Threat Detection and Security Analytics Engine** leveraging **MITRE ATT&CK mapping**, threat scoring, alert correlation, geolocation intelligence, incident tracking, and automated detection workflows for security operations.
- Integrated the **ELK Stack**, Docker, and real-time telemetry pipelines generating attack events every 5 seconds to support **Threat Intelligence**, Security Monitoring, Incident Response, and SOAR Automation through global threat maps and SOC dashboards.

SentinelZero | Cloud Security Monitoring Platform | React, Flask, Python, AWS EC2, Nginx, Cloudflare, Google reCAPTCHA

05/2026

GitHub • [Live Site](#) • [Demo Video](#)

- Built and deployed a cloud security monitoring platform on **AWS EC2** supporting **threat detection, security event monitoring**, incident reporting, and 10+ Flask REST APIs.
- Leveraged the **CIC-IDS2017 intrusion detection dataset** to model malicious traffic patterns, analyze attack behavior, and simulate **SOC-style security monitoring** workflows.
- Implemented **IAM security controls**, authentication protection, secure session management, Google reCAPTCHA, Cloudflare security services, and Nginx reverse proxy to enhance **cloud security posture** and access control.

SecureShield AI | AI-Powered Application Security Monitoring Platform | Python, Flask, React, JavaScript, OWASP Top 10

06/2026

GitHub • [Demo Video](#)

- Developed an **AI-Powered Application Security Platform** capable of detecting and monitoring SQL Injection, Cross-Site Scripting (XSS), and Brute Force attacks through real-time security analysis and attack pattern recognition.
- Engineered automated **Threat Detection** workflows incorporating alert generation, severity classification, attack simulation, IP blocking, security event logging, and incident tracking.
- Built an AI Security Assistant to support **Incident Response** and security investigations by analyzing attack techniques, recommending mitigations, and providing guidance aligned with **OWASP Top 10** security principles.

LEADERSHIP & EXPERIENCE

Cybersecurity Analyst | Blue Team Operations & Threat Detection | CodePath

02/2026 — 05/2026

- Led analysis of 10K+ PCAP packets using **Wireshark**, improving phishing/malicious traffic detection speed by ~50%.
- Monitored endpoints via **Wazuh** + Linux logs, detecting 100+ security events (FIM, SSH/FTP abuse) and reducing unnoticed threats.
- Queried 50K+ **SIEM logs (Splunk)** to identify IOCs and potential breaches, improving detection coverage by ~40%.
- Conducted **incident response** using **NIST framework**, reducing investigation time from hours → minutes with structured workflows.

Cybersecurity Researcher | Electric Vehicle & Grid Security

01/2026 — Present

- Modeled **cyber-physical attacks** on EV-grid systems, identifying vulnerabilities that could impact 1000+ nodes in power distribution networks.
- Built **anomaly detection** models on grid telemetry, improving detection of compromised charging behavior by ~35%.
- Simulated large-scale adversarial attacks on IEEE feeder models, enhancing system resilience and threat preparedness.
- Performed **attack surface analysis + threat intelligence** mapping, strengthening defensive strategies for critical infrastructure.

Data Analyst | Global Career Accelerator

01/2026 — 05/2026

- Built Python data pipelines (Pandas, Jupyter, Plotly) to ingest, clean, and transform **10K+ records per dataset** for real-world analysis.
- Automated data processing and feature extraction, reducing manual data preparation time by ~40%.
- Analyzed multi-domain datasets (SQL + Pandas) using joins, filtering, and aggregations to uncover trends across **5+ datasets**.
- Delivered dashboards and insights (Plotly), translating technical findings into actionable business recommendations.

Vice President of Technology | AWS Club

09/2025 — Present

- Led technical initiatives for 100+ members, organizing **AWS and cloud security** workshops focused on real-world deployment practices.
- Delivered 5+ hands-on sessions covering cloud architecture, **IAM, APIs**, and deployment workflows, increasing technical engagement across members.
- Coordinated large-scale technical events and speaker sessions, boosting participation and attendance by ~30%.
- Mentored students on AWS services, cloud security fundamentals, and project deployment strategies through project-based learning.